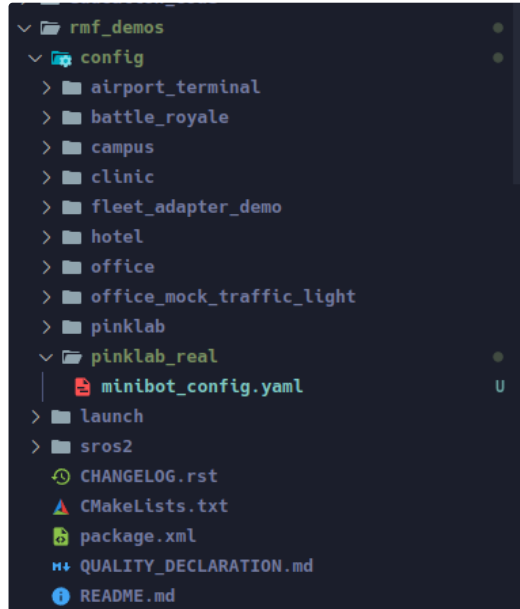


2. Fleet adapter 구성

Fleet adapter를 위한 Config파일을 생성 [🔗](#)

- rmf_demos/config에 pinklab_real폴더를 만들고 그 아래에 config파일 만들기.



RMF시스템 내 Minibot Fleet의 설정 작성. [링크1](#) [🔗](#)

```
rmf_fleet:
  name: "minibot"
  fleet_manager:
    prefix: "http://127.0.0.1:22016"
    ip: "127.0.0.1"
    port: 22016
    user: "some_user"
    password: "some_password"
  limits:
    linear: [0.3, 0.15] # 최대 속도 약 0.3 m/s, 가속도는 0.15 m/s^2
    angular: [1.5, 0.7] # 최대 각속도 약 1.5 rad/s, 가속도는 0.7 rad/s^2
  profile: # Robot profile is modelled as a circle
    footprint: 0.25 # minibot의 반지름 0.25 m
    vicinity: 0.5 # minibot의 안전 반경 0.5 m
  reversible: False # whether robots in this fleet can reverse
  battery_system:
    voltage: 12.0 # V
    capacity: 24.0 # Ahr
    charging_current: 5.0 # A
  mechanical_system:
    mass: 20.0 # kg
    moment_of_inertia: 10.0 # kgm^2
    friction_coefficient: 0.22
  ambient_system:
    power: 20.0 # W
  tool_system:
    power: 0.0 # W
  recharge_threshold: 0.10 # Battery level below which robots in this fleet will not operate
  recharge_soc: 1.0 # Battery level to which robots in this fleet should be charged up to during recharging tasks
  publish_fleet_state: 10.0 # Publish frequency for fleet state, ensure that it is same as robot_state_update_frequency
  account_for_battery_drain: False
  task_capabilities: # Specify the types of RMF Tasks that robots in this fleet are capable of performing
    loop: True
    delivery: False
    clean: False
    finishing_request: "park" # [park, charge, nothing]
  actions: ["teleop"]

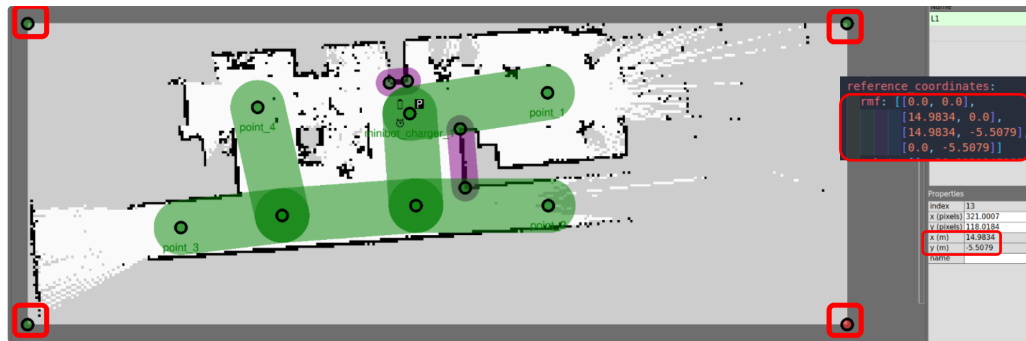
  responsive_wait: True # Should responsive wait be on/off for the whole fleet by default? False if not specified.
```

minibot_config.yaml에 추가로 로봇의 개수, 업데이트 주기, 시작지점, 충전지점을 설정 [링크2](#)

```
# minibot CONFIG =====
robots:
  minibot 1:
    robot_config:
      max_delay: 10.0 # allowed seconds of delay of the current itinerary before it gets interrupted and replanned
    rmf_config:
      robot_state_update_frequency: 1.0
      charger:
        waypoint: "minibot_charger_1"
      start:
        map_name: "L1"
        waypoint: "minibot_charger_1"
        orientation: 0.0 # radians
```

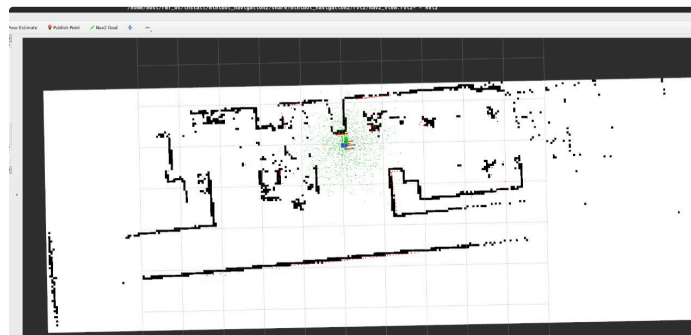
Traffic-editor를 실행해서, 이미지의 4점의 좌표를 minibot_config.yaml에 작성하기.

- 시계방향으로 기입하자.

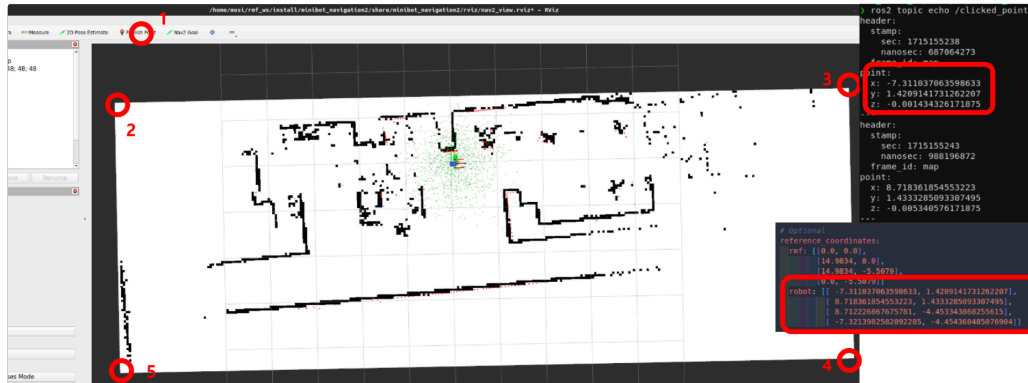


그 다음 모바일 로봇의 Navigation2을 실행하자.

- 1 ros2 launch minibot_bringup bringup_robot.launch.xml # Minibot에서
- 2 ros2 launch minibot_navigation2 bringup_launch.xml \
- 3 map:=\$HOME/rmf_ws/src/rmf_demos/rmf_demos_maps/maps/pinklab_real/pinklab_L1.yaml #PC에서
- 4 rviz2 -d `ros2 pkg prefix minibot\_navigation2`/share/minibot_navigation2/rviz/nav2_view.rviz



터미널을 한개 더 열어서 /clicked_point를 구독한 뒤에, Rviz2에서 네 점을 찍어나온 좌표를 minibot_config에 기입하자. [링크3](#)



기존 RMF실행파일에 Fleet adapter를 실행하는 코드를 추가. [링크4](#)

- RMF를 실행하는 파일은 rmf_demos/rmf_demos/launch/pinklab_real.launch.xml이다.

```
1 <!-- Minibot fleet adapter -->
2 <include file="$(find-pkg-share fleet_adapter_template)/launch/fleet_minibot_adapter.launch.xml">
3   <arg name="use_sim_time" value="$(var use_sim_time)" />
4   <arg name="nav_graph_file"
5     value="$(find-pkg-share rmf_demos_maps)/maps/pinklab_real/nav_graphs/0.yaml" />
6   <arg name="config_file"
7     value="$(find-pkg-share rmf_demos)/config/pinklab_real/minibot_config.yaml" />
8 </include>
```

그리고 Navigation2 정보를 가져오는 Free Fleet Server코드를 추가. [링크5](#)

```
1 <node pkg="free_fleet_server_ros2"
2   exec="free_fleet_server_ros2"
3   name="minibot_fleet_server_node"
4   output="both">
5   <param name="fleet_name" value="minibot" />
6   <param name="fleet_state_topic" value="minibot/fleet_states" />
7   <param name="mode_request_topic" value="minibot/robot_mode_requests" />
8   <param name="path_request_topic" value="minibot/robot_path_requests" />
9   <param name="destination_request_topic" value="minibot/robot_destination_requests" />
10  <param name="dds_domain" value="43" />
11  <param name="dds_robot_state_topic" value="robot_state" />
12  <param name="dds_mode_request_topic" value="mode_request" />
13  <param name="dds_path_request_topic" value="path_request" />
14  <param name="dds_destination_request_topic" value="destination_request" />
15  <param name="update_state_frequency" value="20.0" />
16  <param name="publish_state_frequency" value="2.0" />
17  <param name="translation_x" value="0.0" />
18  <param name="translation_y" value="0.0" />
19  <param name="rotation" value="0.0" />
20  <param name="scale" value="1.0" />
21 </node>
```

이제 Navigation2과 Free Fleet client를 같이 실행하도록 코드를 작성. [링크6](#)

- Navigation2을 시작하는 코드인 pinklab_minibot_robot/minibot_navigation2/launch/bringup_launch.xml 에 Free Fleet client코드를 추가하자.

```
1 <arg name="robot_name" default="minibot_1" />
2 <!-- launch the free fleet client first -->
```

```

3 <node name="minibot_free_fleet_client_node" pkg="free_fleet_client_ros2"
4   exec="free_fleet_client_ros2" output="both">
5   <param name="fleet_name" value="minibot" />
6   <param name="robot_name" value="$(var robot_name)" />
7   <param name="robot_model" value="minibot" />
8   <param name="level_name" value="L1" />
9   <param name="dds_domain" value="43" />
10  <param name="max_dist_to_first_waypoint" value="10.0" />
11  <param name="map_frame" value="map" />
12  <param name="robot_frame" value="base_footprint" />
13  <param name="nav2_server_name" value="/navigate_to_pose" />
14  <param name="use_sim_time" value="False" />
15  <param name="dds_domain" value="43" />
16 </node>

```

빌드 후 Navigation2부터 실행. [↗](#)

```

1 ros2 launch minibot_bringup bringup_robot.launch.xml # Minibot에서
2 ros2 launch minibot_navigation2 bringup_launch.xml \
3   map:=$HOME/rmf_ws/src/rmf_demos/rmf_demos_maps/maps/pinklab_real/pinklab_L1.yaml #PC에서 실행

```

그 다음에 RMF를 실행하면 [↗](#)

```

1 ros2 launch rmf_demos pinklab_real.launch.xml

```



RMF-Panel에서 이동 시키기. [↗](#)

<https://open-rmf.github.io/rmf-panel-js/>

Submit a Task

Select a request type
Loop

Set start time (mins from now)
0

Choose a priority (Default: 0)
0

Schedule a Loop Request

Select start location
minibot_charger_1

Select end location
point_4

Number of Loops
1

SUBMIT REQUEST

로봇이 이동하는 것을 볼 수 있다. 🔗



만약에 로봇 Navigation2의 initial pose가 안맞는다면 navigation2패키지의 nav2_params패키지를 수정하자. [링크](#) 🔗

